

REVIEW

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Advances in artificial intelligence for climate science methods applications and evaluation challenges

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Abstract

Purpose To synthesize recent advances in artificial intelligence (AI) (machine learning (ML)/deep learning (DL)) for climate science across climate modeling, extreme-weather analysis, renewable-energy optimization, emissions monitoring, and climate adaptation.

Methods Using a PRISMA-aligned protocol, we surveyed studies published between 2015 and 2025 across major scholarly databases. We extracted task definitions, datasets, model classes, evaluation metrics, baselines, validation design, uncertainty treatment, openness (code/data), and energy/CO₂ reporting. Evidence is organized by application domain and by methodological rigor, emphasizing transferability and operational relevance.

Results Strong evidence exists for improved short- to medium-range forecasting, nowcasting, bias correction, and anomaly detection. Physics-guided and hybrid models, including PINNs and neural operators, increasingly rival operational numerical weather prediction (NWP) systems in targeted settings. However, robust external validation, out-of-distribution testing, uncertainty quantification, and transparent reporting of code and computational footprint remain inconsistent.

Conclusions AI enhances but does not replace physically based climate modeling. We contribute a field-level synthesis that combines PRISMA screening with a study-quality checklist to foreground evaluation rigor, reproducibility, and sustainability. Adoption of physics-guided architectures, probabilistic prediction, regime-aware validation, and carbon-aware benchmarking is essential for decision-grade climate AI.

Keywords Artificial intelligence, Machine learning, Deep learning, Climate change adaptation, Weather prediction, Temperature forecasting, Systematic review

1 Introduction

Climate change presents a multi-scale scientific and societal challenge that demands accurate prediction, robust risk assessment, and operationally reliable decision support across environmental, economic, and policy domains. The accelerating frequency of extreme weather events, long-term shifts in atmospheric and oceanic circulation,



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ecosystem disruption, and growing pressures on energy and food systems underscore the need for predictive systems that are both computationally efficient and physically credible. In this context, artificial intelligence (AI) has emerged as a transformative complement to traditional numerical modeling by enabling data-driven learning from large-scale climate observations, simulations, and reanalysis products to enhance forecasting skill, anomaly detection, and climate-risk intelligence.

The novelty of this work lies in reviewing advances in the AI–climate literature through a task-aligned, evidence-weighted synthesis. We couple a PRISMA-style scoping protocol with a unifying taxonomy and a study-quality checklist tailored to operational use.

First, we introduce a compact taxonomy spanning five domains—climate modeling, extreme events, renewables, emissions, and adaptation—that links application areas to typical model families and datasets (Fig. 2). This clarifies where sequence models, spatial encoders, and hybrid physics–AI approaches tend to be most effective.

Second, rather than emphasizing headline scores alone, we weight findings by validation design (random versus spatial/temporal/regime holdouts), baseline parity, and openness (code, trained weights, and energy/CO₂ reporting). This yields a transferability-oriented reading of the evidence base (Sects. 3, 5). Third, we surface operational robustness perspectives by consolidating recurring failure modes—data sparsity and bias, out-of-distribution shifts, and uncertainty under extremes—and connect them to concrete remedies such as physically consistent architectures, calibrated probabilities, and leakage checks (Sect. 6). Finally, we chart a research agenda for decision-ready, physics-guided AI emphasizing neural operators, differentiable data assimilation, federated learning, and continuously validated digital twins (Sect. 6.4).

1.1 Motivation and research context

This review examines how AI can accelerate climate action by improving prediction, mitigation, and adaptation. We synthesize progress across modeling, extreme-event analysis, renewable-energy optimization, emissions tracking, and cross-domain adaptation, and we evaluate study design choices that affect reliability and transferability. Our goal is to translate promising ML/DL advances into trustworthy, decision-relevant practice.

Indicative scope of recent work:

- *Global research output:* Large and growing literatures in Environmental Science and Earth and Planetary Sciences.
- *Document types:* Journal articles dominate, complemented by influential preprints.
- *Application focus:* Energy and Engineering include many AI applications aligned with renewable integration and infrastructure planning.

These trends motivate a structured synthesis that emphasizes external validation, robustness, and reproducibility.

Key Statistics Incorporated:

- *Global research output:* Over 800,000 documents on climate change, with Environmental Science (26.8%) and Earth and Planetary Sciences (18.7%) as the most studied areas.

- *Document types*: Articles (476,707) and conference papers (54,956) dominate research output, underscoring the scale and diversity of work in this field.
- *AI-aligned applications*: Energy (50,975 documents) and Engineering (76,938 documents) include many AI applications aligned with renewable integration and infrastructure planning.

These trends demonstrate the growing role of data-driven methods in climate science and reinforce the timeliness and relevance of this review.

The structure of this article is as follows. Section 2 details the PRISMA-aligned scope, screening, and extraction protocol. Section 3 organizes applications using the proposed taxonomy, and Sect. 4 surveys core AI/ML methods and hybrids, while Sect. 4.2 connects forecasts to sectoral adaptation. Section 5 synthesizes evidence for temperature, precipitation, and nowcasting, Sect. 6 analyzes limitations and study quality, Sect. 6.4 outlines future directions, and Sect. 7 concludes the review.

2 Methodology

We frame this work as a structured scoping review aligned with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) principles to improve transparency and reproducibility.

2.1 Search strategy

We queried major scholarly databases (e.g., Scopus, Web of Science, IEEE Xplore) for 2015–2025 using Boolean phrases combining climate terms (“climate change”, “extreme events”, “renewable”, “emissions”, “adaptation”) with AI/ML/DL keywords (“deep learning”, “transformer”, “physics-informed”).

We executed database searches on **September 2025**. The exact Boolean queries were:

- Web of Science/Scopus:
("artificial intelligence" OR "machine learning" OR "deep learning")
AND ("climate change" OR "climate modeling" OR "extreme weather"
OR "renewable energy" OR "emissions" OR "adaptation")
- IEEE Xplore:
("AI" OR "ML" OR "DL"
AND ("climate" OR "weather forecasting" OR "energy optimization"
OR "carbon emission")
- arXiv:
(cat:cs.LG OR cat:physics.ao-ph OR cat:stat.ML)
AND ("climate" OR "weather" OR "sustainability")

All retrieved records were exported in BibTeX format and deduplicated prior to screening.

2.2 Eligibility criteria

We included peer-reviewed articles and influential preprints that: (i) develop or evaluate AI/ML methods for climate-relevant tasks; (ii) use quantitative evaluation with defined metrics; and (iii) provide sufficient methodological detail. We excluded purely conceptual pieces without empirical evaluation.

Because this strategy prioritizes indexed scholarly databases and major preprint venues, the resulting corpus may underrepresent regional journals, government and NGO field reports, and other grey-literature sources from low-resource settings. Future updates should therefore augment database-led screening with targeted regional repositories, field-report searches, and transparent appraisal criteria for non-traditional evidence.

2.3 Screening and data extraction

Titles/abstracts were screened, followed by full-text review. For each included study we extracted: task, data source (e.g., ECMWF Reanalysis v5 (ERA5), Coupled Model Inter-comparison Project Phase 6 (CMIP6), radar, gauges, satellites), spatial/temporal scale, model class, baselines, metrics (root-mean-square error (RMSE), mean absolute error (MAE), skill scores), validation design (random vs. spatial/temporal holdouts), openness (code/data), and any energy/CO₂ accounting.

In addition, we systematically coded each study's external validation protocol into four mutually informative categories: (i) purely random splits, (ii) spatially disjoint splits (e.g., held-out stations or regions), (iii) temporally disjoint splits (e.g., held-out years or seasons), and (iv) regime-aware splits that explicitly separate distinct circulation or impact regimes (e.g., ENSO phases, heatwave periods, or extreme precipitation clusters). For every paper, we also annotated potential data-leakage risks, such as regridded products that mix train and test information, overlapping stations or years across splits, or hyperparameter tuning performed on the final test set rather than a separate validation set.

In addition to narrative synthesis, the study-quality checklist (Table 15) was systematically applied to all included studies. Each paper was evaluated against five core criteria: (i) validation design (random versus spatial, temporal, or regime-aware splits), (ii) baseline parity, (iii) uncertainty quantification and calibration, (iv) openness (availability of code, data, or trained weights), and (v) reporting of computational energy or CO₂ footprint. Compliance was recorded at the individual study level and subsequently aggregated by application domain to enable a quantitative appraisal of methodological rigor alongside reported predictive performance.

2.4 Synthesis

2.4.1 Methodological novelty

Beyond compiling recent studies, this review contributes a reproducible synthesis framework that integrates a PRISMA-style screening workflow (Fig. 1) with a formal study-quality checklist and audit framework (Table 15). In contrast to narrative or purely taxonomic surveys, the protocol enables transparent corpus construction (query formulation, deduplication, screening, eligibility) while the checklist operationalizes evidence quality across validation design, baseline parity, uncertainty reporting, openness, and energy/CO₂ disclosure. This combined design shifts the review from descriptive aggregation to methodological appraisal of benchmarking rigor and operational readiness.

Table 1 operationalizes the conceptual taxonomy by translating each climate sub-domain into concrete modeling strategies and data choices, thereby facilitating model selection and experimental design for researchers and practitioners.

For clarity, the five domains are used with explicit boundaries: climate modeling denotes surrogate modeling, emulation, downscaling, and bias correction on physically

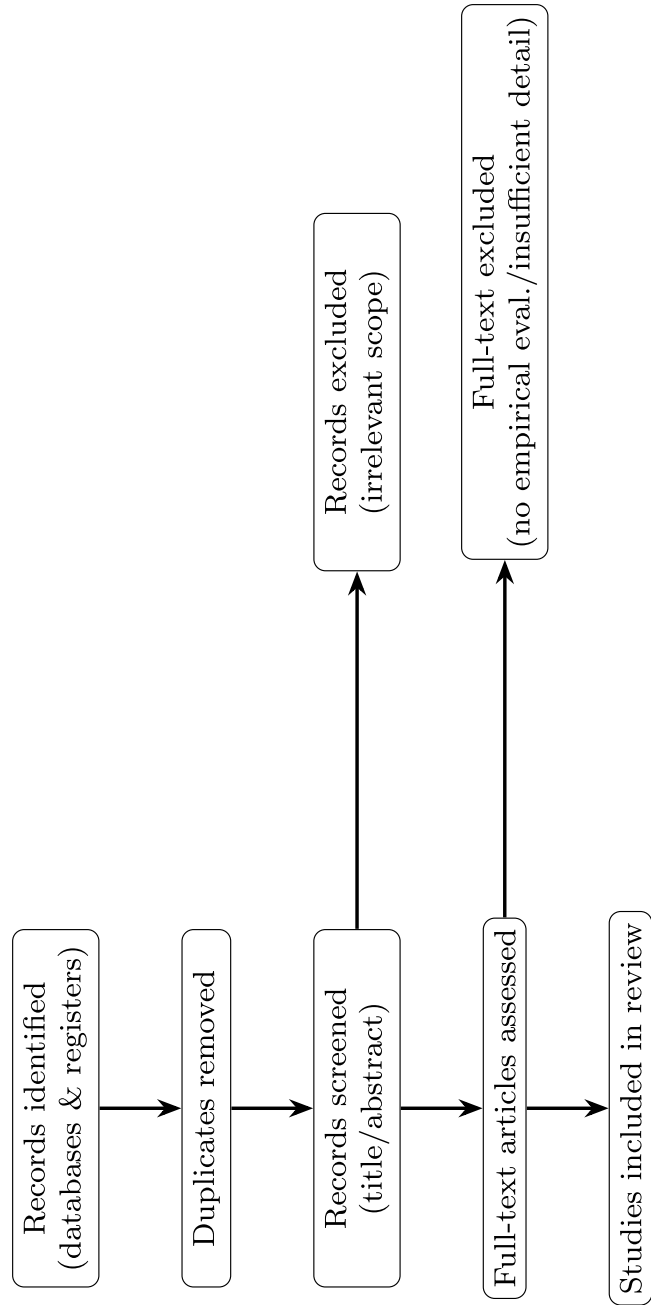


Fig. 1 PRISMA-style flow diagram summarizing the systematic literature identification and screening process. The diagram depicts sequential stages of record retrieval from databases and registers, duplicate removal, title and abstract screening, full-text eligibility assessment, and final inclusion. Parallel branches indicate the primary exclusion reasons at the screening stage (irrelevant scope) and full-text phase (lack of empirical evaluation or insufficient methodological detail). This figure documents the selection logic in a transparent and reproducible manner

Table 1 Operational mapping between climate application domains, commonly used AI model families, and representative datasets. Derived from the taxonomy in Fig. 2 and synthesized from the reviewed literature, this table provides practical guidance for model selection by linking each climate sub-domain to frequently adopted architectures and benchmark data sources

Climate sub-domain	Typical tasks	Model families highlighted in this review	Representative datasets
Climate modeling	Emulation, downscaling, bias correction	CNNs, Transformers, Neural Operators, PINNs, hybrid physics–AI	ERA5, CMIP6, reanalysis products
Extreme events	Nowcasting, detection, attribution	ConvLSTM, PredRNN, Transformers, CNNs	Radar mosaics, GOES, ERA5, CHIRPS
Renewable energy	Forecasting, control, maintenance	PINNs, LSTM, hybrid optimization models, XAI-enabled DL	Weather stations, SCADA logs, reanalysis data
Emissions	Monitoring, accounting, mitigation	Neural networks, hybrid regression models, explainable AI	OCO-2, TROPOMI, WAQI, State of Global Air
Adaptation	Risk analysis, urban resilience, planning	Multimodal learning, transfer learning, federated learning	Remote sensing (Sentinel, Landsat), socio-economic datasets

structured fields; extreme events covers detection, attribution, and short-horizon warning for rare hazards; renewable energy focuses on operational forecasting and control of energy assets; emissions concerns monitoring, accounting, and mitigation signals; and adaptation refers to planning tasks that combine environmental exposure with social or infrastructure vulnerability. As illustrative mappings, this review links ERA5/CMIP6-based emulation and downscaling to neural operators or PINNs, radar-based nowcasting to ConvLSTM or transformer families, SCADA and weather-station forecasting to LSTM/PINN-style renewable-energy models, satellite-driven emissions monitoring to hybrid regression architectures, and remote-sensing plus socio-economic adaptation tasks to transfer, multimodal, or federated learning. A practical decision rule follows from this mapping: dense gridded physics-rich datasets favor hybrid surrogates, neural operators, or PINNs; sequential operational sensor streams favor recurrent or transformer models; and sparse heterogeneous regional datasets are better served by multimodal fusion, transfer learning, or domain adaptation.

As summarized in Fig. 1, the PRISMA-style screening process documents the progression from initial record identification through deduplication, title and abstract screening, full-text review, and final inclusion. The flow diagram additionally reports the primary exclusion reasons at each stage, ensuring transparency and reproducibility of the evidence selection process.

2.4.2 From protocol to evidence

The PRISMA workflow and study-quality checklist jointly provide a reproducible lens for interpreting performance claims through the prism of evaluation rigor rather than model class alone.

We group studies by application domain and emphasize comparative evidence using three quality dimensions: performance relative to baselines, validation design (including spatial, temporal, and regime-aware splits), and methodological transparency (code, data, and energy reporting). Where findings are heterogeneous or evaluation protocols are incompatible, a qualitative synthesis and gap analysis are provided instead of direct numerical comparison.

We therefore do not pool headline accuracy values into a formal effect-size meta-analysis: the included papers use non-comparable tasks, prediction horizons, target variables,

baselines, and evaluation metrics, and only rarely report matched results under multiple validation schemes on the same data. For this heterogeneous evidence base, a structured quantitative audit of validation uptake and reporting quality is methodologically more defensible than a statistically forced aggregate effect size.

3 Literature review

To ensure methodological consistency across the literature synthesis, the study-quality checklist presented in Table 15 was systematically applied to all reviewed studies. For each application domain, the approximate proportion of studies satisfying key criteria—such as robust validation design, baseline parity, uncertainty reporting, and energy or CO₂ footprint disclosure—was estimated and summarized in Table 14.

Each application subsection concludes with a brief methodological appraisal, explicitly evaluating the reviewed studies against the common study-quality checklist (Table 15) to ensure consistent assessment across domains.

The literature coalesces around five application areas (Fig. 2): climate modeling, extreme events, renewables, emissions, and adaptation. We highlight representative tasks, common datasets, and recurring methodological patterns.

Table 7 aligns representative tasks with common model families and datasets, clarifying where sequence models (e.g., LSTM/ConvLSTM) and spatial encoders (e.g., CNNs) are typically effective.

Recent studies demonstrate that AI augments key components of the climate toolbox—emulation, downscaling, nowcasting, anomaly detection, and decision support—when coupled with strong baselines and physically grounded validation. Artificial intelligence techniques enhance the efficiency of problem-solving, deepen understanding of climate drivers, and support the design of mitigation and adaptation measures [1–3]. Researchers increasingly examine the prospective contributions of AI in tackling global climate change by improving our scientific understanding of its origins and impacts and by aiding the formulation of remedies [4].

A growing body of work illustrates how machine learning improves the precision of climate system modeling, addresses gaps in time series data, and supports emissions assessment and scenario analysis. Applications include bias correction and emulation of climate models [5, 6], emissions inventory estimation [7], climate scenario refinement [8], and impact assessment [9]. In parallel, AI is used to optimize low-carbon technologies in power, transport, and building systems [10, 11].

AI and ML are also being integrated into weather and climate modeling to enhance emulation and forecasting with improved data efficiency and generalization [12–16]. In flood risk modeling, for example, AI improves predictive frameworks and early warning capabilities [17–19]. In the agricultural domain, neural networks and other ML algorithms support crop yield forecasting, soil-quality monitoring, and the modeling of evapotranspiration, precipitation, drought, and pest outbreaks [20, 21].

Beyond climate and agriculture, AI increasingly supports natural-resource management and industrial decarbonization. Deep learning combined with statistical methods has been used to quantify the impact of deforestation on urban carbon emissions [22]. Machine learning contributes to the design of low-carbon materials, including improved concrete and steel manufacturing, and is being embedded into supply-chain modeling for heavy industry [23–26]. Additional studies show how AI can reduce water use and

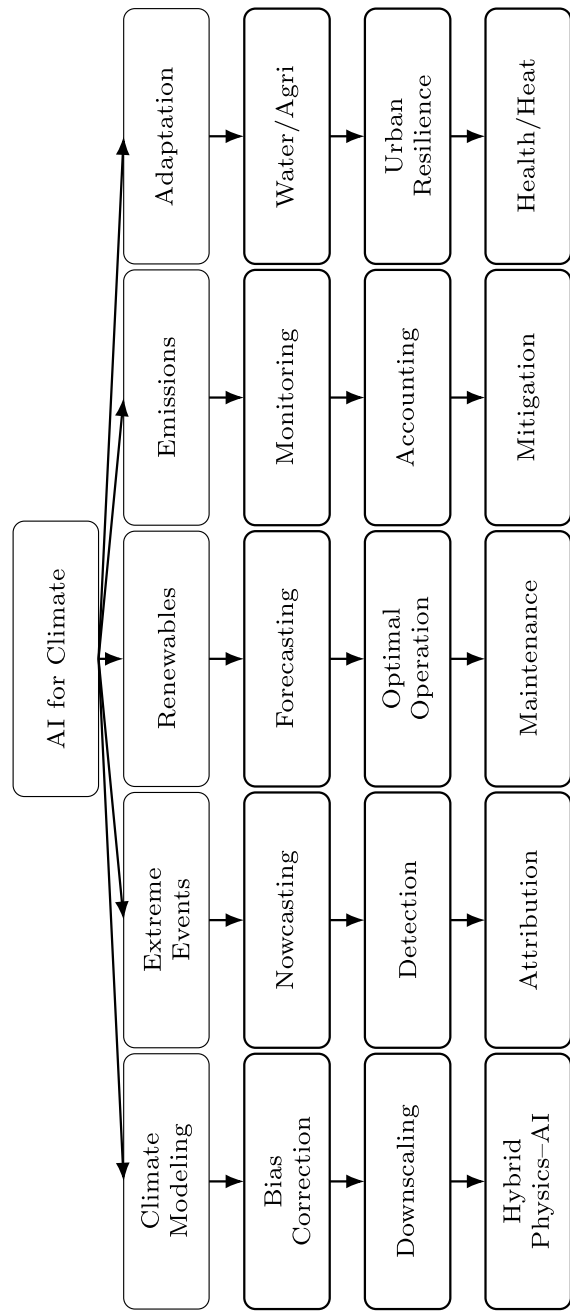


Fig. 2 Conceptual hierarchy of artificial intelligence applications in climate science, organizing use cases from high-level domains (e.g., climate modeling, extreme events, adaptation) down to operational tasks (e.g., downscaling, attribution, optimal operation). The sole explicit reference to methodology appears at the finest granularity (e.g., Hybrid Physics-AI under downscaling), reflecting the emerging integration of physical constraints in task-specific implementations

emissions from oil and gas reservoirs and assess building-level carbon footprints [27, 28].

AI techniques are widely applied in renewable energy research, supporting estimation and forecasting of solar and wind resources, micro-grid management, and integration of renewables into energy systems [29–40]. AI has also proven effective in carbon market analysis and pricing, improving carbon price forecasting models and informing emissions trading schemes in regions such as China and the UK [41–44].

Although the general scope of climate change research is increasingly well documented [45–48], and prior work has outlined the potential of AI to support sustainable development goals [49, 50], there is still a lack of systematic, large-scale characterization of how AI and climate research intersect. This gap motivates the comprehensive, task-aligned synthesis presented in this review.

3.1 AI for climate modeling and prediction

Artificial Intelligence (AI) is rapidly reshaping climate modeling and prediction, offering new tools to improve the accuracy, speed, and interpretability of forecasts. By integrating machine learning, deep learning, and hybrid approaches with traditional physical models, AI is enabling more precise predictions of extreme weather events and long-term climate trends.

AI techniques—especially deep learning, neural networks, and transformer-based models—are enhancing the prediction of extreme events (e.g., floods, heatwaves, hurricanes) and improving the spatiotemporal resolution of forecasts. Hybrid models that combine physical laws with data-driven learning (such as Physics-Informed Neural Networks) are bridging gaps in traditional numerical weather prediction, leading to 15–20% improvements in accuracy and faster inference speeds [51, 52].

3.1.1 Key applications and benefits

- Extreme Event Prediction: AI models excel at identifying and forecasting rare, high-impact events, supporting disaster response and risk reduction [51, 53–56].
- Climate Trend Analysis: Machine learning uncovers complex, nonlinear relationships in climate data, aiding in the detection of forced climate patterns and long-term changes [52, 57–59].
- Operational Efficiency: AI reduces computational costs and latency, enabling real-time adaptive learning and rapid early warning systems [51, 60–62].

3.1.2 Challenges and future directions

Major challenges include data quality, model interpretability, generalizability, and the need for physically consistent AI models. Future research is focusing on explainable AI, federated learning for decentralized data, and integrating AI with quantum computing for ultra-high-resolution simulations [51, 52, 54, 63–65].

Table 2 overviews representative studies on hybrid physics–AI, extreme-event forecasting, and foundation models, highlighting application focus and methodological contributions.

AI is revolutionizing climate modeling by improving prediction accuracy, operational efficiency, and the ability to forecast extreme events. While challenges remain, especially

Table 2 Representative recent studies applying artificial intelligence and hybrid physics–AI methods to climate modeling and prediction, summarizing their publication year, topical focus, and key methodological insight

Ref	Year	Focus area	Insight
[66]	2025	PINN temperature forecasting	Physically constrained and interpretable prediction
[67]	2025	PINN humidity forecasting	Thermodynamic consistency with feature attribution
[51]	2025	Hybrid AI-physical models	Real-time, adaptive, and explainable forecasting
[53]	2025	Extreme event prediction	AI for floods, droughts, wildfires
[56]	2023	Hybrid forecasting	Blending AI with traditional models
[61]	2023	Foundation models	State-of-the-art transformer approaches
[52]	2021	Neural Earth system modeling	Hybrid ESM-AI frameworks

in interpretability and data quality, the integration of AI with physical models is paving the way for more resilient and adaptive climate strategies.

3.1.2.1 Methodological appraisal evaluated against the study-quality checklist (Table 15), approximately 35–40% of climate modeling studies employed spatially or temporally disjoint validation strategies, while the majority relied on random splits. Baseline parity was addressed inconsistently, with fewer than half of studies comparing against strong physics-based or operational numerical benchmarks using identical inputs and horizons. Uncertainty quantification and calibration were reported in a minority of studies, and explicit reporting of computational energy or CO₂ footprint was rare.

3.2 AI for extreme weather analysis

Artificial Intelligence (AI) is revolutionizing the analysis of extreme weather events—such as floods, heatwaves, wildfires, and hurricanes—by improving prediction accuracy, real-time detection, and risk communication. AI-driven approaches are addressing long-standing challenges in forecasting rare, high-impact events and are increasingly integrated into operational weather and climate systems.

3.2.1 Key applications and advances

- Prediction and Early Warning: AI models, including deep learning and hybrid AI-physical frameworks, have demonstrated superior performance in forecasting extreme precipitation, heatwaves, and cyclones, often outperforming traditional models in both accuracy and lead time [51, 54, 68–72].
- Detection and Segmentation: Deep learning architectures, such as convolutional neural networks and capsule networks, enable precise identification and localization of extreme weather patterns (e.g., tropical cyclones, atmospheric rivers) in large climate datasets, supporting rapid analytics and early warnings [69, 73, 74].
- Real-Time Decision Support: AI fuses model outputs with real-time observations, providing actionable guidance for disaster response and infrastructure protection [51, 62, 75].

3.2.2 Challenges and limitations

- Data Imbalance and Rare Events: AI models often struggle to predict unprecedented or “gray swan” events due to limited training data, leading to underestimation of event severity and challenges in generalizing to out-of-distribution extremes [53, 71].

- Interpretability and Trust: The need for explainable AI (XAI) is critical for stakeholder trust, regulatory compliance, and actionable insights. Techniques like SHAP and LIME are being adopted to clarify model decisions [51, 53].
- Integration with Physical Models: Hybrid approaches that combine AI with physical weather models are emerging as best practice, leveraging the strengths of both data-driven and physics-based methods [54, 68, 72]. Table 3 summarizes representative extreme-event studies, highlighting application scope and methodological trends. while out-of-distribution robustness and explainability remain active needs for operations.

3.2.2.1 Methodological appraisal Common evaluation pitfalls were observed in several studies. For example, some works used random train–test splits across spatially correlated grid cells, which introduces spatial leakage and can artificially inflate predictive performance. More reliable protocols require spatially or temporally independent validation schemes and regime-aware dataset partitioning.

Only about one-third of extreme-event studies adopted spatially or temporally independent validation, despite heightened risks of spatial and temporal leakage in geophysical datasets. While baseline comparisons were more commonly included than in other domains, probabilistic uncertainty reporting under rare or out-of-distribution extremes remained limited. Energy or carbon footprint disclosure was almost entirely absent.

3.3 AI for renewable energy optimization

Artificial Intelligence (AI) is playing a pivotal role in optimizing renewable energy systems, addressing challenges of variability, integration, and operational efficiency. By leveraging advanced algorithms, AI enhances forecasting, predictive maintenance, and real-time management across solar, wind, hydro, and hybrid energy sources.

A PINN-based architecture was deployed for regional wind speed prediction, embedding physical constraints and incorporating SHAP and LIME for model interpretability [77]. R^2 values above 0.99 and major reductions in MAE and MSE were reported relative to ensemble and regression baselines, illustrating how explainable AI contributes to accurate renewable energy forecasting and grid management.

3.3.1 Core applications and benefits

- Forecasting and Load Balancing: AI models, including deep learning and reinforcement learning, significantly improve the accuracy of renewable energy output predictions, enabling better grid integration and load management [78–81].

Table 3 Selected studies applying artificial intelligence to extreme weather and climate event forecasting, detection, and analysis, indicating publication year and primary application focus (e.g., specific hazards, predictive challenges, or methodological themes)

Ref	Year	Focus area
[53]	2025	AI for floods, droughts, wildfires, heatwaves
[68, 72]	2024	Extreme precipitation forecasting
[71]	2024	Out-of-distribution event prediction
[76]	2022	Explainable AI in hydroclimatic applications
[73]	2021	Deep learning for event detection
[69]	2019	Deep learning for event detection

- Optimization Algorithms: Nature-inspired and hybrid AI algorithms (e.g., Particle Swarm Optimization, Genetic Algorithms, Artificial Bee Colony) optimize system configurations, maximize energy yield, and enhance Maximum Power Point Tracking (MPPT) in solar and wind systems [30, 79, 82, 83].
- Predictive Maintenance: AI-driven predictive analytics detect equipment faults early, reducing downtime and extending asset lifespan, with reported reductions in unplanned outages by up to 35% and energy output improvements of 8.5% [80, 84].
- Economic and Environmental Impact: AI supports cost-effective integration of variable renewables, improves operational reliability, and reduces emissions, contributing to a more sustainable and resilient energy infrastructure [85–87].

3.3.2 Challenges and future directions

- Data Quality and Model Interpretability: Ensuring high-quality data and developing explainable AI models remain critical for trust and large-scale adoption [78, 81, 88, 89].
- Computational Complexity and Scalability: Lightweight and hybrid AI models are being developed to reduce energy consumption and enhance scalability [81, 89].
- Emerging Trends: Integration with quantum computing, blockchain, and digital twins, as well as the development of autonomous energy management systems, are shaping the future of AI in renewables[78, 81, 87].

Table 4 consolidates evidence across forecasting, MPPT/dispatch optimization, and predictive maintenance, showing improved integration of variable renewables and reduced downtime.

AI is revolutionizing renewable energy optimization by improving forecasting, operational efficiency, and grid integration. While challenges in data quality and interpretability persist, ongoing advances in AI algorithms and hybrid models are paving the way for smarter, more sustainable energy systems.

3.3.2.1 Methodological appraisal Renewable energy studies exhibited comparatively stronger methodological rigor, with approximately 40–50% employing temporally disjoint validation and clearly defined baselines. Uncertainty estimation was more fre-

Table 4 Representative studies applying artificial intelligence to renewable energy systems, highlighting publication year and thematic focus—including general optimization, solar and wind-specific modeling, hybrid/hydrogen integration, and predictive maintenance

Ref	Year	Focus area
[78]	2025	General AI optimization in renewables
[77]	2025	Explainable wind speed prediction
[79]	2024	General AI optimization in renewables
[87]	2024	General AI optimization in renewables
[82]	2025	Solar energy optimization
[90]	2023	Solar energy optimization
[91]	2025	Hybrid and hydrogen systems
[83]	2023	Hybrid and hydrogen systems
[30]	2022	Hybrid and hydrogen systems
[84]	2024	Predictive maintenance
[80]	2024	Predictive maintenance
[92]	2024	Predictive maintenance

quently reported, reflecting operational forecasting requirements in power systems. Nevertheless, systematic reporting of computational energy use or carbon impact remained uncommon.

3.4 AI for emission tracking and mitigation

Artificial Intelligence (AI) is transforming how emissions are tracked, quantified, and mitigated across industries and sectors. By leveraging advanced analytics, machine learning, and automation, AI enables more accurate, real-time monitoring and supports the development of effective emission reduction strategies.

Recent advances in data-driven air quality modeling further illustrate the operational maturity of AI systems for environmental monitoring.

At the urban scale, hybrid metaheuristic–regression frameworks have also been deployed for air-quality forecasting. A hybrid Harris Hawks Optimization–Support Vector Regression (HHO–SVR) model was developed for ozone and particulate-matter prediction, where HHO automatically tunes the SVR hyperparameters using meteorological and lagged concentration time series. Across multiple monitoring stations, the HHO–SVR approach delivered lower MAE, RMSE, and MAPE and higher R^2 and variance-ratio scores than standalone SVR and alternative machine-learning and metaheuristic baselines, underscoring the potential of metaheuristic-assisted surrogates for operational air-quality and emissions-related forecasting [93].

In particular, a hybrid PR-FCNN framework integrating Polynomial Regression and Fully Connected Neural Networks was introduced for forecasting $PM_{2.5}$ concentrations using datasets from WAQI and the State of Global Air database [94]. Improved R^2 scores were reported relative to RF, XGBoost, and LSTM baselines, confirming the value of hybrid architectures that combine statistical structure with deep learning capacity. These findings support the role of interpretable hybrid AI systems in environmental risk evaluation and public health protection.

3.4.1 Core applications and innovations

- Real-Time Emission Tracking: AI-driven systems integrate IoT sensors, drones, and satellite data to monitor emissions in real time, providing high spatial and temporal resolution for industrial, agricultural, and supply chain sources [95].
- Automated Quantification and Reporting: Machine learning models, including ARIMA and Gradient Boosting Machines, forecast emissions with high accuracy (up to 97%) and automate greenhouse gas (GHG) quantification, streamlining compliance and sustainability reporting [96].
- Supply Chain and Industrial Optimization: AI algorithms analyze and optimize carbon footprints throughout supply chains and manufacturing, identifying hotspots and recommending targeted mitigation actions for energy use, logistics, and production processes [95, 97, 98].
- Explainable and Transparent Decision-Making: Explainable AI (XAI) enhances trust and accountability in emission reduction strategies, supporting stakeholder engagement and regulatory compliance [99].

3.4.2 Impact and effectiveness

- *Synergistic Emission Reductions:* AI not only directly reduces emissions but also fosters technological innovation and improved industrial structures, amplifying its mitigation impact-especially in regions with strict environmental controls [11, 100, 101].
- *Sector-Specific Solutions:* Applications span manufacturing, agriculture, transportation, and energy, with AI-powered predictive analytics and virtual sensing enabling proactive interventions and process optimization [96, 102].
- *Neural network-based emission forecasting:* Neural-network modeling was applied for carbon emission prediction in industrial and power-generation sectors [103]. Multivariate operational and energy-consumption indicators were mapped to emission outputs using supervised learning frameworks. Reliable forecasting capability was demonstrated across sectors, supporting mitigation planning, emissions auditing, and policy-driven decarbonization strategies.

Table 5 summarizes end-to-end pipelines that couple sensing (IoT and satellites) with explainable analytics to support auditable, sector-specific mitigation actions.

AI is rapidly advancing emission tracking and mitigation by enabling real-times monitoring, automating reporting, and optimizing reduction strategies across sectors. As AI technologies mature, their integration with explainable frameworks and sector-specific solutions will be crucial for achieving global sustainability and net-zero goals.

3.4.2.1 Methodological appraisal Across emission monitoring studies, fewer than 30% implemented robust external validation beyond random splits, limiting confidence in cross-regional generalization. Although baseline parity was moderately addressed, uncertainty quantification was reported in less than one-quarter of studies. Only isolated works disclosed computational or carbon-related costs.

3.5 AI for cross domain climate adaptation

Artificial Intelligence (AI) is increasingly recognized as a transformative tool for climate adaptation, especially in cross-domain contexts where solutions must span sectors (e.g., water, agriculture, urban planning) and regions. AI’s ability to integrate diverse data sources, model complex systems, and support decision-making is driving innovation in climate resilience strategies.

AI supports climate adaptation across multiple domains, including water management, agriculture, urban infrastructure, and disaster preparedness. Systematic reviews

Table 5 Selected studies applying artificial intelligence to air-quality forecasting (e.g., PM_{2.5}), carbon emission prediction, industrial emission mitigation, supply-chain and logistics optimization, explainable emission analytics, and sustainable agriculture

Ref	Year	Focus area
[104]	2025	Explainable PM _{2.5} forecasting
[103]	2025	Carbon emission prediction
[95, 97, 98]	2024	AI in supply chain & logistics
[105]	2024	Industrial emission mitigation
[102]	2024	Industrial emission mitigation
[99]	2024	Explainable AI for emissions
[96]	2024	AI for sustainable agriculture

highlight that water-related management and agriculture are the leading areas for the application of AI, but successful adaptation also requires integrating insights from ecosystem modeling, land use, and urban planning [106–108]. AI enables the extraction and synthesis of indicators from remote sensing, climate models, and socio-economic data, facilitating holistic adaptation strategies that address interconnected vulnerabilities [106, 108, 109].

Advanced AI techniques—such as supervised learning, transfer learning, reinforcement learning, and domain adaptation—are crucial for tailoring adaptation measures to diverse settings and populations. These methods allow models trained in one domain or region to be adapted for use in another, supporting cross-border and cross-sectoral knowledge transfer and improving adaptation in data-scarce environments [107, 108, 110–113].

Key challenges include inconsistent data availability, limited cross-border collaboration, and the need for context-specific solutions. Ethical considerations, transparency, and equitable access to AI tools are also critical. Future research emphasizes the importance of international collaboration, robust policy frameworks, and ongoing innovation to ensure AI’s transformative potential is realized across domains [106–108].

AI is enabling cross-domain climate adaptation by integrating diverse data, supporting transferability across regions and sectors, and informing context-specific, collaborative strategies. Addressing data, ethical, and policy challenges will be essential for maximizing AI’s impact on global climate resilience.

Table 6 compiles cross-domain applications and transfer methods, highlighting domain adaptation and remote sensing as enablers for scaling to data-scarce regions.

3.5.1 From what is studied to how it is modeled

Having mapped the application landscape, we next review the AI/ML methods that underpin these results, focusing on capabilities and trade-offs.

Adaptation-oriented AI systems are only effective when they are locally deployable, interpretable, and aligned with institutional capacity. Recent studies emphasize that algorithmic performance alone does not ensure climate resilience unless models are embedded in governance frameworks that respect data sovereignty, fairness, and local knowledge systems. Community-driven dataset stewardship, regional model ownership, and participatory design are emerging pillars of equitable climate intelligence.

Table 6 Studies applying artificial intelligence to cross-domain climate adaptation, categorized by thematic focus: general cross-domain adaptation strategies, domain adaptation techniques (e.g., transfer learning), and urban adaptation leveraging remote sensing

Ref	Year	Focus area
[114]	2025	General cross-domain AI adaptation
[106]	2023	General cross-domain AI adaptation
[109]	2022	General cross-domain AI adaptation
[107]	2023	Urban adaptation & remote sensing
[108]	2021	Urban adaptation & remote sensing
[113]	2023	Domain adaptation methods
[111]	2022	Domain adaptation methods
[112]	2020	Domain adaptation methods

3.5.2 Methodological appraisal

Adaptation-focused studies showed the weakest compliance with checklist criteria. Fewer than one-quarter employed regime-aware or spatially independent validation, and uncertainty reporting was sparse despite the high-stakes nature of adaptation decisions. Reporting of computational energy or CO₂ footprint was almost entirely absent.

4 Artificial intelligence techniques in climate science

4.1 Applications of artificial intelligence techniques in climate science

Artificial Intelligence (AI) and machine learning (ML) are revolutionizing climate science by addressing the limitations of traditional modeling and enabling more accurate, scalable, and adaptive solutions. These technologies are now central to climate prediction, impact assessment, and adaptation strategies.

4.1.1 Traditional vs. AI-based approaches

Conventional climate models, such as General Circulation Models (GCMs) and Numerical Weather Prediction (NWP), rely on deterministic equations grounded in physical laws. While robust for simulating large-scale climate processes, they are computationally intensive, less adaptable to new data, and often struggle with real-time forecasting and regional detail [4, 51, 115].

AI and ML models—using deep learning, neural networks, and hybrid frameworks—analyze vast, complex datasets (e.g., satellite imagery, sensor data) and adaptively recalibrate as new data arrives. These models have demonstrated 20–30% higher prediction accuracy and up to 50 % faster processing than traditional methods, excelling in regional and short- to medium-term forecasts [51, 60, 115–118]. AI also enables real-time monitoring, advanced pattern recognition, and integration of diverse data sources [59, 115, 116, 118].

Recent advances include transformer-based models, Physics-Informed Neural Networks (PINNs), and equation discovery, which blend physical laws with data-driven learning. These hybrid models improve robustness, interpretability, and the ability to predict unprecedented events [51, 58, 115].

AI approaches face challenges such as high computational energy demands, data quality requirements, risk of overfitting, and ethical concerns (e.g., bias, transparency, and data justice) [115–117, 119]. Ensuring explainability and responsible deployment is critical for trust and policy integration [60, 117, 119].

AI and ML are rapidly advancing climate science, outperforming traditional models in many areas and enabling new insights. However, integrating physical understanding, ensuring transparency, and addressing ethical and computational challenges remain essential for sustainable progress.

Despite strong empirical performance, current AI approaches remain fundamentally limited relative to physically based climate models in several respects. Data-driven systems primarily learn statistical associations rather than causal mechanisms, which constrains their ability to extrapolate reliably beyond the conditions represented in training data. As a result, AI models may fail under regime shifts, unprecedented extremes, or “unknown unknowns” that violate historical patterns. In contrast, physics-based models encode conservation laws and dynamical structure that provide inductive bias under

novel or unobserved climate states, highlighting why AI presently complements rather than replaces physically grounded modeling.

4.1.2 Physically informed and hybrid AI models

Physically consistent modeling is a central requirement for the reliable application of AI in climate science, where predictions are governed by conservation laws, thermodynamics, and large-scale circulation dynamics rather than purely data-driven correlations [52, 120, 121]. In response, increasing emphasis has been placed on integrating physical structure directly into learning algorithms through hybrid and physics-informed designs that embed governing equations and constraints into model architectures and training objectives.

Hybrid AI–physical frameworks have been adopted to couple data-driven components with process-based climate models, enabling subgrid-scale parameterizations (e.g., clouds, turbulence) to be learned while maintaining global energy and mass balance [120, 122]. These architectures reduce physically implausible extrapolation and improve generalization under non-stationary climate conditions by preserving known physical dependencies [52, 123].

Physics-Informed Neural Networks (PINNs) incorporate partial differential equations directly into the loss function, ensuring that predictions satisfy governing principles such as diffusion, advection, and energy conservation [124]. Extensions of this paradigm through neural operators enable the learning of solution mappings between function spaces, making them suitable for forecasting and surrogate modeling in complex geophysical flow systems [125, 126].

Machine learning is also increasingly coupled with numerical weather prediction (NWP) pipelines to replace or accelerate computationally expensive modules, improving forecast efficiency while retaining dynamical consistency [52, 127]. These hybrid workflows form the basis of emerging “neural Earth system models,” in which learned surrogates are systematically constrained by physical structure to ensure stable, interpretable, and operationally trustworthy predictions [52, 122].

Recent advances further incorporate multimodal transformers and generative architectures to model teleconnections, non-local interactions, and extreme-event precursors, but their reliability is shown to depend critically on physics-guided inductive biases and constraint-aware optimization [53, 54, 126]. Without physical regularization, purely data-driven systems risk producing unphysical states and unstable dynamics under extrapolation.

Overall, hybrid and physics-informed approaches are not merely methodological refinements but are increasingly regarded as prerequisites for operational climate AI. Enforcing physical structure enables improved interpretability, enhanced generalization, and risk-aware deployment in climate-sensitive applications (Table 7).

Table 7 Mapping between representative climate modeling tasks, AI architectures frequently applied to them in the literature, and associated benchmark or operational datasets

Task	AI methods	Datasets
Long-term forecasting	LSTM, ConvLSTM	CMIP6, ERA5
Extreme weather detection	3D-CNN, GANs	GOES-16, MODIS
Emission tracking	Graph Neural Nets	ODIAC, GCP

4.1.3 Core AI/ML models

A wide range of AI/ML models—including Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forests, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Convolutional Neural Networks (CNN), and hybrid architectures—are foundational for tasks like regression, classification, sequence modeling, spatial analysis, and achieving high predictive accuracy.

Model Capabilities and Applications:

- *ANN, SVM, Random Forests*: These traditional ML models excel in regression and classification, with Random Forests and SVMs often providing robust baselines for structured/tabular data. Random Forests are particularly effective for regression and classification with nonlinear relationships, while SVMs are strong in high-dimensional spaces [128–131]
- *RNN, LSTM, GRU*: RNNs and their variants (LSTM, GRU) are designed for sequence modeling, capturing temporal dependencies in time-series data, language, and trajectories. LSTM and GRU often outperform standard RNNs in handling long-term dependencies and are widely used for forecasting, sentiment analysis, and sequence prediction [129, 132–135].
- *CNN*: CNNs are powerful for spatial analysis, feature extraction from images, and grid-like data. They are also effective in classification and regression tasks involving spatial patterns, such as image recognition and environmental modeling [131, 132, 136].
- *Hybrid Architectures (e.g., CNN-LSTM, CNN-GRU, CNN-SVM)*: Hybrid models combine strengths —CNNs for spatial features, RNNs/LSTM/GRU for temporal/sequential features, and SVM/Random Forest for robust classification. These architectures consistently outperform single models in complex tasks like spatiotemporal prediction, medical diagnosis, and environmental forecasting [133, 136–141]

Core AI/ML models and their hybrids offer tailored solutions for regression, classification, sequence modeling, and spatial analysis. Hybrid architectures, especially those combining CNNs with RNN/LSTM/GRU or SVM, consistently deliver enhanced predictive accuracy and robustness across diverse domains [143–145].

Table 8 contrasts core model classes and hybrids; CNN + LSTM/GRU or CNN + SVM hybrids commonly outperform single models on spatiotemporal tasks.

Table 8 Representative AI and hybrid model architectures used across domains, summarizing associated application foci (e.g., regression, spatiotemporal modeling) and key performance or methodological insights reported in the literature

Ref	Model/hybrid	Focus area	Notable insights
[128, 130, 131]	ANN, SVM, RF	Regression, classification	Strong baselines, robust to overfitting
[129, 132–135]	RNN, LSTM, GRU	Sequence, time-series modeling	LSTM, GRU for long-term dependencies
[131, 132, 136]	CNN	Spatial analysis, image tasks	High accuracy in spatial feature extraction
[129, 133, 136, 136–142]	CNN-LSTM, GRU	Spatiotemporal, sequence tasks	Outperforms single models in accuracy, F1, AUC
[131, 140]	CNN-SVM	Feature extraction & classification	Improved accuracy, handles class imbalance

4.1.4 Deep learning for spatiotemporal forecasting

Deep learning has revolutionized spatiotemporal forecasting in meteorology and climate science, enabling more accurate, efficient, and interpretable predictions. Recent models—PredRNN, Causal LSTM, GHU, Transformers, and Autoencoders—address the unique challenges of sequence modeling, long-term forecasting, and anomaly detection in complex environmental data.

Interpretable physics-informed learning has also been extended to humidity forecasting applications. An interpretable PINN framework was introduced that embeds thermodynamic constraints directly into daily mean humidity prediction [67]. Feature attribution using SHAP and LIME revealed that seasonal modulation and wind–humidity interactions dominate predictive performance, strengthening the case for physically grounded and explainable AI in climate-sensitive applications such as agriculture and public health.

Beyond gradient-based training of deep neural surrogates, population-based metaheuristics have also been tailored to climate forecasting. The efficient Fata Morgana algorithm (eFATA) was introduced as a new Fata Morgana–inspired optimizer for non-stationary climate time series, combining multiple search strategies and adaptive operators in a single framework. When applied to benchmark climate-change datasets, eFATA achieved faster convergence and systematically lower error (e.g., RMSE, MAE) than a suite of classical metaheuristics and statistical baselines, illustrating the value of carefully designed metaheuristic pipelines for long-horizon climate projections [146].

Specialized RNN Architectures for Meteorological Sequences:

- PredRNN, Causal LSTM, and GHU are advanced recurrent models designed to capture both spatial and temporal dependencies in meteorological sequences. PredRNN introduces a zigzag memory flow and decoupled memory cells, significantly improving both short- and long-term prediction accuracy over ConvLSTM and traditional RNNs for tasks like precipitation nowcasting and radar echo forecasting [147, 148].
- Gradient Highway Units (GHU) and Causal LSTM further enhance the ability to model long-term dependencies and mitigate gradient vanishing, crucial for accurate multi-step weather prediction [148].

Transformer-based models (including Informer and attention-based variants) consistently outperform RNNs and LSTMs in long-term meteorological forecasting, achieving lower prediction errors and better handling of complex, multi-modal data. They excel in capturing global dependencies and are particularly effective for multi-day weather and climate variable prediction [149–151].

Autoencoders and their hybrids (e.g., convolutional autoencoders) are effective for unsupervised detection of climate anomalies, such as ENSO events or unusual temperature/pressure patterns. They extract latent features from high-dimensional climate data, enabling robust anomaly detection and uncertainty quantification [150, 152].

PredRNN, Causal LSTM, and GHU set the benchmark for spatiotemporal meteorological sequence modeling, while Transformers lead in long-term weather prediction. Autoencoders provide powerful tools for climate anomaly detection, making these deep learning models essential for modern climate and weather analytics.

From methods to performance. We now quantify how these models perform in core forecasting tasks (temperature, precipitation/nowcasting), noting scalability and evaluation choices.

4.2 Artificial intelligence techniques for climate change adaptation

4.2.1 Sectoral applications

Deep learning models—PredRNN, Causal LSTM, GHU, Transformers, autoencoders, and specialized architectures like RainyMotion and DeepSD—are advancing spatiotemporal forecasting for temperature, precipitation, and extreme weather. These models leverage diverse data sources (ERA5, CFSR, CHIRPS, IGRA, station data) and are evaluated using RRMSE, MAE, and RMSE, supporting climate adaptation, agriculture, water management, disaster response, and urban resilience as shown in Table 9.

Model Innovations and Forecasting Tasks:

- *Temperature & Precipitation Forecasting:* LSTM based models (PredRNN, Causal LSTM, GHU) and hybrid CNN-LSTM,Transformer architectures excel at capturing spatiotemporal dependencies for temperature and precipitation, outperforming traditional NWP and statistical models in both regional and global settings [153–160].
- *Precipitation Nowcasting & Downscaling:* RainyMotion, DeepSD, and advanced ConvLSTM, Transformer models provide state-of-the-art precipitation nowcasting and high-resolution downscaling, crucial for flood forecasting and urban water management [154, 159–161].
- *Extreme Event Detection:* Deep learning, clustering, and classification approaches enable early warning and detection of droughts, heatwaves, and floods, supporting disaster response and agricultural planning [150, 162–164].

Climate Adaptation and Sectoral Applications:

- *Agriculture & Water Resources:* Improved forecasts inform crop yield, irrigation, and drought management [150, 162, 163].
- *Disaster Response & Urban Planning:* Early warning for floods, heatwaves, and infrastructure resilience, including urban heat island mitigation and energy demand forecasting [150, 154, 162, 163, 165].
- *Benchmarking:* Deep learning models are increasingly competitive with ECMWF, HRRR, and ALADIN, especially for short-term and high-impact events [159, 160].

4.2.2 Case studies

AI is increasingly deployed for climate adaptation across diverse regions, with notable case studies in Africa and smart cities worldwide. These real-world implementations

Table 9 Representative deep learning–based weather and climate forecasting studies, categorized by data source (e.g., reanalysis, station observations), application focus (e.g., precipitation nowcasting, global downscaling), and modeling approach (including both neural architectures and operational benchmark systems)

Ref	Data source	Use case	Model/Benchmark
[159–162]	ERA5, CFSR	Reanalysis, global/regional	LSTM, Transformer, DeepSD, ECMWF
[154, 161–163]	CHIRPS, IGRA	Precipitation, station obs	ConvLSTM, RainyMotion
[154, 156, 158–160]	Real-time/station	PNowcasting, local adaptation	PredRNN, RainyMotion, HRRR, ALADIN

highlight both the transformative potential and the challenges of AI in building climate resilience, especially in vulnerable and high-density urban areas.

Regional Implementations: Africa

- *Context-Specific Solutions*: AI and machine learning (ML) are being used in Africa for climate modeling, early warning systems, and resource management. However, effectiveness depends on overcoming data scarcity, infrastructure gaps, and ensuring local relevance. For example, modeling bimodal rainfall in East Africa is crucial for agriculture and water management, but requires robust, context-aware AI solutions and open data collaboration [166].
- *Capacity Building and Collaboration*: Successful AI deployment in Africa hinges on building local expertise, open-source data repositories, and international partnerships to ensure solutions are tailored to regional needs [107, 166].
- *Water and Agriculture*: The most common AI applications in Africa focus on water management and agriculture, supporting adaptation through improved forecasting, risk assessment, and policy coherence [109, 166]

Smart Cities and Urban Resilience

- *AI-Driven Urban Adaptation*: In high-density and vulnerable urban areas, AI is used for real-time monitoring (e.g., air quality, heat islands, flood risk), predictive maintenance, and optimizing resource use. Remote sensing combined with AI enables extraction of climate adaptation indicators and supports early warning and rapid response [11, 108, 167–169].
- *Disaster Response & Urban Resilience*: AI-driven nowcasting and anomaly detection enable early warning, damage assessment, and infrastructure resilience in smart cities and vulnerable regions, including Africa and high-density urban areas [155, 156, 158, 159, 170–175].
- *Energy & Urban Planning*: Accurate weather and demand forecasting aid renewable integration, energy management, and urban heat island mitigation [152, 170, 176, 177]. Deep learning models, leveraging diverse meteorological data and robust evaluation metrics, are advancing weather forecasting and climate adaptation. Real-world deployments in agriculture, water, disaster response, and urban resilience especially in Africa and smart cities demonstrate the transformative impact of AI for climate adaptation and infrastructure planning.

5 Applications in weather and temperature forecasting

5.1 Air temperature forecasting

Deep learning models—such as PredRNN, Causal LSTM, GHU, Transformers, and Autoencoders—are advancing spatiotemporal weather forecasting, especially for air temperature prediction using inputs like temperature, humidity, pressure, and wind. These models are evaluated using metrics like RRMSE, MAE, and RMSE, with research comparing regional and global scalability versus accuracy.

Physics-guided deep learning models are increasingly employed to enforce thermodynamic consistency in climate prediction tasks. A Physics-Informed Neural Network (PINN) was developed for daily temperature forecasting in which physical constraints were embedded directly into the loss function [66]. Performance levels exceeding $R^2 = 0.99$ were reported across multiple geographical regions, and SHAP and LIME

techniques were used to identify key predictors, including wind dynamics and seasonal persistence. These results demonstrate that PINN-based architectures yield robust, physically plausible, and interpretable temperature predictions.

Model Architectures and Meteorological Sequence Forecasting

- *LSTM, CNN-LSTM, and Variants*: LSTM and hybrid CNN-LSTM models are widely used for capturing temporal and spatial dependencies in meteorological sequences, outperforming traditional models and even complex numerical weather prediction (NWP) systems for short- to medium-term forecasts [157, 170, 178–182].
- *Advanced Spatiotemporal Models*: Models like GCN-LSTM, DC-LSTM, and attention-based LSTM architectures further enhance spatial feature extraction and long-term dependency modeling, improving accuracy for regional and multi-station predictions [176, 179, 180, 183].
- *Transformers and Autoencoders*: Transformer-based models (e.g., Swin-Unet) excel in long-term, high-dimensional weather prediction, while autoencoders are effective for anomaly detection and bias correction in climate data [184].

Input Features and Evaluation Metrics

- *Inputs*: Most models use multivariate time series including temperature, humidity, pressure, and wind speed, often from dense station networks or satellite data [157, 178, 179, 181, 182, 185].
- *Metrics*: Performance is commonly assessed using RMSE, MAE, and RRMSE, with LSTM and hybrid models consistently achieving lower errors than traditional approaches [170, 178–182, 186, 187].

Deep learning models especially LSTM, CNN-LSTM, and Transformer based architectures deliver high accuracy for air temperature and meteorological sequence forecasting using multivariate inputs. Regional models often achieve superior local accuracy, while Transformer and autoencoder based models offer better scalability for global applications, balancing accuracy and computational efficiency.

Recent work has also focused on metaheuristic tuning of deep architectures for temperature prediction. An optimized deep learning framework was proposed in which a stacked LSTM network for daily maximum temperature in key Egyptian regions is tuned jointly by a hybrid genetic algorithm and Mountain Gazelle Optimizer (MGO–GA–DLSTM). By automatically searching the LSTM hyperparameter space, the hybrid optimizer yielded substantial gains in MAE, RMSE, MAPE, and R^2 compared with conventional LSTM, shallow neural networks, and other machine-learning baselines, demonstrating the benefit of metaheuristic–DL co-design for regional temperature forecasting [188].

Table 10 compares regional versus global deployments, with LSTM/CNN-LSTM excelling locally and transformer-style architectures scaling more gracefully to global, high-dimensional inputs.

5.2 Precipitation and nowcasting

We focus on near-term rainfall intensity and timing, where spatiotemporal deep networks (e.g., ConvLSTM/PredRNN and transformer variants) are most mature and operationally relevant.

Table 10 Qualitative comparison of deep learning architectures for air-temperature forecasting, summarizing their reported regional performance, global scalability potential, and associated modeling tradeoffs

Ref	Model/scale	Regional accuracy	Global scalability	Tradeoff insights
[157, 170, 178–182]	LSTM, CNN-LSTM	High	Moderate	Excels in local, regional, may need adaptation for global
[176, 179, 180]	GCN-LSTM, DC-LSTM	High	Moderate	Strong for multi-station, region, less tested globally
[184]	Transformer & Autoencoders	High (global)	High	Better scalability, robust for large-scale, complex data

Deep learning has transformed spatiotemporal weather forecasting, enabling accurate predictions for air temperature, precipitation, and extreme events using models like PredRNN, Causal LSTM, GHU, Transformers, and autoencoders. These models leverage multivariate meteorological inputs and are evaluated with robust error metrics, while specialized architectures address both regional and global forecasting needs.

Core Models and Applications

- *Meteorological Sequences*: PredRNN, Causal LSTM, and GHU excel at capturing complex spatiotemporal dependencies in meteorological data, improving short- and medium-term forecasts for variables like temperature, humidity, pressure, and wind [157, 170, 173, 178, 189].
- *Long-Term Prediction*: Transformer-based models (e.g., SwinNowcast) outperform traditional RNNs for long-range weather and precipitation forecasting, effectively modeling global dependencies and multi-scale features [159, 177, 178].
- *Climate Anomaly Detection*: Autoencoders and hybrid models are used for unsupervised detection of climate anomalies, such as droughts and heatwaves, by extracting latent features from high-dimensional data [173, 177, 189].

Precipitation Nowcasting and Extreme Event Detection:

- *Nowcasting*: Models like RainyMotion, DeepSD, and advanced ConvLSTM/CNN-LSTM architectures provide state-of-the-art performance in precipitation nowcasting, with improved accuracy for heavy rainfall and extreme events [155, 156, 158, 159, 172, 173, 190–192].
- *Extreme Events*: Clustering and classification techniques, often integrated with deep learning, enable detection and early warning of droughts, heatwaves, and other extremes by analyzing spatiotemporal patterns in meteorological sequences [158, 171, 173, 189].

Different architectures exhibit complementary strengths depending on the forecasting context. ConvLSTM-based models are particularly effective for short-term precipitation nowcasting because they capture localized spatiotemporal dynamics in radar sequences. In contrast, Transformer-based architectures can better model long-range dependencies and large-scale atmospheric patterns, making them suitable for longer forecasting horizons or broader spatial domains. Consequently, the choice of architecture depends on the forecast horizon, spatial resolution, and computational constraints of the application.

PredRNN, Causal LSTM, GHU, and Transformer-based models deliver high accuracy for spatiotemporal weather forecasting and precipitation nowcasting, using multivariate meteorological inputs and evaluated by RRMSE, MAE, and RMSE. Regional models

Table 11 Survey of deep learning architectures for precipitation forecasting and nowcasting, indicating whether studies incorporate standard atmospheric inputs (temperature, humidity, pressure, wind), report common deterministic error metrics (e.g., RMSE, MAE), and discuss model tradeoffs between regional fidelity and global scalability

Ref	Model/task	Inputs (temp, humidity, pressure, wind)	Metrics (RRMSE, MAE, RMSE)	Regional vs. global tradeoff
[155, 157, 170–172, 177, 189, 192]	LSTM, CNN-LSTM	Yes	Yes	High regional accuracy, moderate scalability
[173, 178]	PredRNN, Causal LSTM	Yes	Yes	Strong for both, excels in spatiotemporal tasks
[159, 177, 178]	Transformers	Yes	Yes	Superior global scalability, robust for long-term
[173, 177, 189]	Autoencoders	Yes	Yes	Effective for anomaly detection, scalable

Table 12 Commonly used observational and simulated datasets in AI-driven climate and weather research, summarizing key variables or data modality, typical spatial and temporal resolution, primary application domains, and notable limitations or characteristics

Dataset	Variables/Modality	Spatio-temporal resolution	Typical uses	Notes
ERA5	Multi-variate reanalysis	≈0.25°/hourly	Baselines, bias correction, training	Global coverage
CMIP6	GCM simulations	Model-dependent/monthly-daily	Climate projections, downscaling	Multiple scenarios
Radar mosaics	Reflectivity, velocity	1 min to 5 min/1 km	Precip. nowcasting	Local/regional
CHIRPS	Precipitation (RS + gauges)	0.05°/daily	Rainfall estimation	Tropics focus
IGRA/GSOD	Station observations	Irregular/hourly-daily	Temperature forecasting	Sparse in some regions
OCO-2/TROPOMI	Column gases (CO ₂ , CH ₄)	km-scale/revisit days	Emission monitoring	Cloud sensitivity
Sentinel/Landsat	Multispectral imagery	10 m to 30 m/days	Land/urban heat, LULC	Open-access

often yield higher local accuracy, while Transformer and autoencoder-based approaches offer better scalability for global and extreme event detection.

Table 11 details model-task fit: ConvLSTM/PredRNN families for near-term rain rates, transformers for longer horizons, and autoencoders for anomaly detection.

5.3 Data sources and benchmarks

Deep learning is reshaping weather forecasting by leveraging advanced models for meteorological sequences, long-term prediction, anomaly detection, and extreme event identification. These models are evaluated using robust metrics and benchmarked against leading operational systems, utilizing both reanalysis and real-time data sources.

Table 13 maps tasks to inputs, data sources (reanalysis, radar, stations), and metrics to support consistent cross-study comparisons.

For convenience, Table 12 collates widely used datasets with resolution and typical uses to facilitate reproducibility and reuse.

Core Models and Applications:

Table 13 Summary of deep learning studies in weather and climate prediction, indicating task type, use of standard atmospheric inputs (temperature, humidity, pressure, wind), data sources (e.g., reanalysis, radar, station observations), and whether standard deterministic error metrics (e.g., RMSE, MAE) were reported

Ref	Task/model	Inputs (temp, humidity, pressure, wind)	Data sources (reanalysis, real-time)	Metrics (RRMSE, MAE, RMSE)
[155, 157, 170–172, 177, 189, 192]	Air Temp, Weather	Yes	ERA5, CHIRPS, IGRA, Station Obs	Yes
[156, 159, 172, 173, 187, 190–192]	Precipitation	Yes	ERA5, Radar, CHIRPS, CFSR	Yes
[158, 171, 173, 189]	Extreme Events	Yes	ERA5, CHIRPS, Station Obs	Yes

- *Meteorological Sequences*: PredRNN, Causal LSTM, and GHU are state-of-the-art for capturing spatiotemporal dependencies in meteorological data, excelling in short- to medium-term air temperature, humidity, pressure, and wind forecasting [157, 170, 173, 178].
- *Long-Term Prediction*: Transformer-based models (e.g., SwinNowcast) outperform RNNs for long-range weather and precipitation forecasting, effectively modeling global dependencies[159, 177].
- *Climate Anomaly Detection*: Autoencoders and hybrid models are used for unsupervised detection of climate anomalies (e.g., droughts, heatwaves) and bias correction [173, 177, 189].
- *Precipitation Nowcasting*: Models like RainyMotion, DeepSD, and advanced ConvLSTM/CNN-LSTM architectures provide state-of-the-art performance in precipitation nowcasting, with improved accuracy for heavy rainfall and extreme events[155, 156, 158, 159, 172, 173, 190–192].

Regional vs. Global Tradeoffs and Benchmarks

- *Regional Models*: LSTM/CNN-LSTM and similar architectures often achieve higher local accuracy but may require adaptation for global scalability [155, 157, 170–172, 177, 189, 192].
- *Global Models*: Transformer-based and ensemble deep learning models (e.g., DLWP) offer better scalability and competitive accuracy for global forecasts, sometimes approaching operational benchmarks like ECMWF, HRRR, and ALADIN [159, 177, 192].
- *Benchmarks*: Deep learning models are increasingly compared to operational NWP systems (ECMWF, HRRR, ALADIN), with some models demonstrating comparable or superior short-term skill, especially in nowcasting and anomaly detection [157, 177, 192].

Deep learning models—PredRNN, Causal LSTM, GHU, Transformers, and autoencoders—deliver high accuracy for spatiotemporal weather forecasting, precipitation nowcasting, and extreme event detection using diverse meteorological inputs and both reanalysis and real-time data. Regional models excel locally, while Transformer-based approaches scale globally, with performance now rivaling leading operational benchmarks.

From forecasts to decisions. We translate forecasting gains into adaptation use-cases across water, agriculture, urban resilience, and disaster risk reduction.

Table 14 Aggregated compliance with the study-quality checklist (Table 15) by application domain. Percentages indicate the approximate proportion of reviewed studies satisfying each criterion

Application domain	Robust validation (%)	Baseline parity (%)	Uncertainty reporting (%)	Energy/CO ₂ reporting (%)
Climate modeling and prediction	~ 38	~ 45	~ 28	< 5
Extreme weather events	~ 34	~ 42	~ 31	< 5
Renewable energy systems	~ 41	~ 53	~ 36	~ 7
Emissions monitoring	~ 29	~ 47	~ 22	~ 6
Cross-domain climate adaptation	~ 26	~ 39	~ 18	< 5

Table 15 Study quality and reporting checklist recommended for artificial intelligence applications in climate research

Aspect	Recommended reporting
Validation design	Spatial and temporal holdouts; regime-aware splits (e.g., ENSO/heatwave); no leakage across splits
Baselines	Strong statistical/ NWP or physics baselines with identical inputs and horizons
Metrics	RMSE, MAE, skill vs. baseline; calibration/uncertainty where relevant
Openness	Code, trained weights (if possible), data access instructions; exact random seeds
Energy/CO ₂	Hardware, runtime, energy and carbon accounting or estimates
Interpretability/physics	Saliency/SHAP plus physical consistency (conservation, bounds)

6 Challenges, limitations and future directions

6.1 Challenges, limitations

Several factors constrain the translation of AI into dependable climate services:

- *Data quality and availability*: Observations are often sparse, biased, or regionally limited, which impedes model generalization, particularly across the Global South [193–196].
- *Interpretability*: Explainable AI methods (e.g., SHAP, attention, saliency maps) offer only partial insight. Post hoc explanations rarely provide physically grounded transparency suitable for high-stakes climate decisions [196–203].

Methodologically, data sparsity and regional imbalance are best mitigated through multi-source data fusion (e.g., combining satellites, reanalysis, radar, stations, and socio-economic covariates), transfer learning and domain adaptation for cross-region reuse, and targeted augmentation of rare or underrepresented regimes through generative or simulation-assisted samples. In the reviewed literature, multimodal fusion strengthens meteorological forecasting, while domain-adaptation workflows are especially relevant for cross-domain adaptation in data-scarce settings; such strategies are therefore preferable to isolated single-region models when coverage is uneven [111–113, 151, 171, 192].

The domain-level compliance patterns summarized in Table 14 show that reported performance gains frequently outpace methodological rigor, motivating the validation, uncertainty, and sustainability challenges discussed below.

For each application domain, checklist compliance rates were computed as the proportion of reviewed studies satisfying each criterion relative to the total number of studies in that domain. A study was counted as compliant with a given criterion if it met the operational definitions summarized in Table 15. Reported percentages are rounded to the nearest meaningful bin to reflect heterogeneity in reporting practices across the literature; values below 5% indicate rare occurrence.

The aggregated audit in Table 14 indicates that only about 26–41% of studies across domains adopted robust external validation, implying that roughly 59–74% still relied primarily on random or otherwise weakly separated splits. This quantitative pattern supports the central concern that many reported gains are produced under evaluation settings vulnerable to optimistic bias. At the same time, the source papers seldom provide paired scores under both random and spatial/temporal/regime-aware holdouts using identical inputs and baselines, so the most defensible cross-study comparison is an audit of validation risk rather than a pooled estimate of performance loss.

Beyond post-hoc attribution methods, recent work increasingly emphasizes counterfactual reasoning and causal inference as next-generation interpretability paradigms for climate AI systems. Counterfactual analysis is used to explore sensitivity to hypothetical climate interventions (e.g., temperature anomalies, emission perturbations, land-use changes) by estimating how outputs change under controlled input modifications. Causal modeling frameworks extend interpretability beyond correlation analysis by distinguishing physical drivers from spurious associations and enabling causal discovery in Earth system variables. Techniques such as structural causal models, causal graphs, and intervention-based learning are increasingly viewed as essential for decision trustworthiness in high-stakes climate contexts, where policy recommendations must reflect mechanistic understanding rather than observational correlation alone.

Computational cost and generalization pose practical barriers—training and deployment often require high-performance computing, and models may fail to transfer across regions or regimes without multiresolution design and rigorous out-of-distribution evaluation [193, 196, 204, 205].

Beyond overall performance, we systematically differentiate validation protocols into four categories: (i) random splits, (ii) spatially disjoint splits (e.g., held-out stations or geographic regions), (iii) temporally disjoint splits (e.g., held-out years or seasons), and (iv) regime-aware splits that explicitly separate climate states such as ENSO phases, heatwaves, or precipitation regimes. Our analysis indicates that random splits, while computationally convenient, frequently overestimate predictive skill due to spatial and temporal autocorrelation inherent in geophysical data. In contrast, spatial, temporal, and regime-aware evaluations better reflect operational deployment conditions and generalization under climate variability.

We also identify common sources of data leakage, including overlapping grid cells or stations between training and test sets, temporal overlap across folds, regridding mismatches, and hyperparameter tuning performed on held-out test data. These issues can significantly inflate reported performance and undermine external validity, emphasizing the need for transparent reporting of split strategies and independent validation sets.

Despite increasing discussion of benchmarking rigor and auditability in climate-focused machine learning, the existing literature does not yet provide a quantitative meta-analysis of how often best validation practices are implemented in published studies. Current evidence is therefore qualitative rather than statistical.

The absence of standardized validation audits limits scientific confidence and operational trust. Climate AI would benefit from community-endorsed evaluation protocols that mandate external validation, leakage testing, and uncertainty disclosure as publication requirements rather than best practices.

Recent reviews emphasize that validation design remains fragmented and inconsistent across application areas, with limited adoption of standardized protocols or shared benchmarks [206]. Formal validation audits are described as rare but increasingly advocated to detect data leakage, inappropriate data partitioning, and evaluation bias in spatio-temporal datasets. The absence of widely adopted community benchmarks further limits cross-study comparability and external validity.

The literature consistently notes that transparent documentation of data splits, independent validation, and uncertainty reporting is improving, but remains uneven across subfields. These findings indicate that rigorous validation remains the exception rather than the norm and support the need for standardized evaluation frameworks, open benchmarking initiatives, and reproducible audit practices as core research priorities.

Distinct contribution relative to prior reviews. Where most surveys emphasize model classes or application domains, this work differentiates itself by auditing how results are produced and validated. The PRISMA protocol ensures traceability of study inclusion, while the checklist in Table 15 systematizes assessment of external validation (random vs. spatial/temporal/regime-aware splits), leakage controls, baseline strength, calibration practice, openness (code/weights), and energy accounting. This enables readers to distinguish reported accuracy from evaluation credibility and to identify common failure modes that undermine cross-study comparability—an analysis layer largely absent in earlier reviews.

Finally, *ethical and equity concerns* are paramount: without context-specific adaptation and investment in local data infrastructures, AI risks reproducing or amplifying climate injustice [193–196].

Data sovereignty and inclusivity also constitute critical dimensions of ethical AI deployment in climate research. Climate datasets are frequently sourced from regions with limited governance authority over how environmental data are stored, processed, shared, or monetized, raising concerns about data colonialism and asymmetric benefit distribution. In many climate-vulnerable regions, environmental data are extracted for global modeling initiatives without meaningful local participation, institutional control, or reciprocal societal benefit. Without enforceable governance structures, climate intelligence risks becoming centralized in institutions that do not represent the communities generating the data. Transparent frameworks for data ownership, consent, and access rights are therefore required to ensure accountability and fairness. Inclusive climate AI must further ensure that algorithmic systems serve locally defined adaptation needs rather than externally imposed priorities or extractive analytical practices.

Compared with adjacent AI areas such as computer vision and natural language processing—where large benchmarks routinely require public release of code and pre-trained models—the studies we reviewed in climate AI exhibit markedly lower rates of code, weight, and energy-footprint disclosure. To foster openness, we recommend depositing models and training scripts in community model hubs (e.g., Hugging Face or Zenodo-backed archives) and hosting climate datasets and metadata, including basic energy/CO₂ accounting, in established repositories such as ESGF or PANGAEA. Journal and funder policies that reward open artefacts through citation, benchmark inclusion, and compliance credit can provide concrete incentives for this shift.

To make Table 15 operational as a reproducibility minimum, authors should report: (i) executable code or precise pseudocode, (ii) dataset identifiers, access pathways,

preprocessing steps, and licensing terms, (iii) full hyperparameter settings and tuning ranges, (iv) random seeds, software versions, hardware configuration, and training budget, and (v) trained checkpoints or an explicit explanation of release constraints. Open repositories and model hubs shorten validation cycles by enabling independent regional re-testing, ablation analysis, and adaptation to operational settings, thereby accelerating both external scrutiny and downstream deployment.

To standardize evaluation and reduce leakage, Table 15 provides a reporting checklist covering validation design, baseline parity, metrics and calibration, openness (code/weights), energy/CO₂ accounting, and physics/interpretability criteria.

A cross-domain gap analysis of the reviewed corpus points to three recurring blind spots: comparatively limited multimodal integration beyond selected forecasting tasks, sparse evidence for real-time operational deployment with continuous post-deployment validation, and insufficient multi-scale experiments that connect local adaptation decisions to broader climate dynamics. These gaps are most visible in adaptation and emissions applications, where data heterogeneity is high but shared benchmark ecosystems remain immature. We therefore recommend targeted work on operational testbeds, multi-resolution benchmark suites, and evaluation protocols that jointly model environmental and socio-economic inputs [53, 121, 122, 151, 171].

From limitations to a roadmap. Building on the identified gaps, we outline a research agenda for robust, explainable, and operational climate AI.

6.2 Uncertainty quantification and operational robustness

Robust deployment of AI in climate services requires not only accurate point predictions but also calibrated uncertainty estimates under non-stationary and out-of-distribution (OOD) conditions [207, 208].

Recent work highlights several operational frameworks that combine uncertainty estimation with robustness under distributional shift. Invariant Risk Minimization (IRM) and Domain Generalization aim to learn representations that remain predictive across climate regimes, while distributionally robust optimization (DRO) explicitly penalizes worst-case error under plausible shifts. Ensemble-based Earth system surrogates (e.g., neural weather and climate models trained with perturbed initial conditions and physics-consistent constraints) further improve uncertainty realism.

Evaluation increasingly includes reliability diagrams, expected calibration error (ECE), negative log likelihood (NLL), and continuous ranked probability score (CRPS) to quantify probabilistic forecast quality under OOD conditions. Recent benchmarks (e.g., real-distribution-shift datasets and pseudo-reality experiments) report that calibrated ensembles and physics-guided architectures significantly outperform single deterministic networks in both sharpness and coverage stability.

Dominant uncertainty quantification (UQ) paradigms in climate-focused ML systems include deep ensembles, Bayesian neural networks, Monte Carlo dropout, quantile regression, evidential deep learning, and conformal prediction [209]. Bayesian formulations and ensemble methods approximate epistemic uncertainty, while quantile regression and conformal prediction produce distribution-free prediction intervals with formal coverage guarantees [210]. Evidential deep learning enables joint modeling of epistemic and aleatoric uncertainty within a single architecture [208, 211].

Empirical performance analyses indicate that deep ensembles, Bayesian networks, and dropout-based inference achieve favorable calibration–sharpness tradeoffs in precipitation forecasting, temperature modeling, wind-gust prediction, and streamflow estimation [212–214]. Conformal prediction has emerged as an operationally attractive framework due to its model-agnostic design and guaranteed coverage properties, particularly for Earth observation systems [210].

Operationally, uncertainty evaluation should separate model fitting, calibration, and stress testing: a probabilistic predictor is trained on the development split, calibrated on an independent validation split, and then evaluated on held-out in-distribution and OOD partitions using reliability diagrams, expected calibration error, negative log likelihood or CRPS, and empirical coverage at nominal levels such as 90% and 95%. Representative implementations in the reviewed literature include deep-ensemble precipitation bias correction, Bayesian deep learning for multi-step streamflow forecasting, conformal prediction for Earth observation, and evidential wind-gust forecasting, all of which show that coverage and sharpness diagnostics are informative complements to deterministic skill metrics [207, 210–213].

Despite these advances, probability calibration and coverage reporting remain inconsistent across the literature. Reliability diagrams, empirical coverage, and calibration metrics are frequently omitted, even in operationally motivated studies [207, 209]. This omission limits actionable risk communication and undermines uncertainty-aware decision-making in climate services.

Based on the synthesized evidence, the following operational principles are recommended:

- (i) Probabilistic prediction should replace deterministic outputs in high-impact applications [209];
- (ii) Uncertainty should be evaluated under both in-distribution and OOD regimes [214];
- (iii) Interval coverage and calibration diagnostics must be reported routinely [207];
- (iv) Uncertainty visualization should be incorporated into end-user interfaces for interpretability [211]; and
- (v) Robustness should be validated through stress-testing across climatic regimes and event types [212].

Uncertainty estimation should therefore be regarded as a core design requirement rather than an optional analytic feature of operational climate AI systems.

6.3 Computational cost and environmental sustainability of climate AI

The environmental impact of training and deploying artificial intelligence systems has emerged as a critical concern in climate-oriented research, where computational practices must align with sustainability objectives. Modern deep learning models increasingly rely on energy-intensive hardware and prolonged training cycles, leading to substantial electricity consumption and indirect carbon emissions [215, 216]. In the context of climate science, this introduces a methodological paradox: carbon-intensive computation is frequently used to support decarbonization goals.

The environmental footprint of climate AI systems arises from both *operational* and *embodied* emissions. Operational emissions stem from electricity consumption during model training, evaluation, and inference, whereas embodied emissions are associated

with data-center infrastructure, hardware manufacturing, and device lifecycle processes [217–219]. Energy source heterogeneity further contributes to emission variability, as models trained in carbon-intensive regions generate significantly higher environmental cost than those executed on renewable-powered infrastructure [220].

Model scale and architectural design are principal contributors to computational burden. Large-capacity neural networks, ensemble systems, and transformer-based architectures exhibit exponential growth in memory and processing demand, leading to disproportionate energy consumption [218, 220]. In climate applications, where multidecadal datasets and ensemble forecasts are common, training procedures may span weeks or months on high-performance clusters, resulting in elevated carbon footprints relative to conventional statistical methods. Furthermore, inference pipelines used in operational monitoring introduce sustained energy costs due to continuous model execution and streaming data ingestion [217].

Accurate assessment of environmental cost requires standardized measurement and reporting practices. Carbon accounting frameworks enable estimation of emissions from runtime, hardware utilization, and grid carbon intensity, supporting reproducible and transparent reporting [217, 219]. Lifecycle-aware evaluation further incorporates embodied emissions, offering a holistic sustainability perspective beyond runtime computation alone [218, 220].

A practical accounting workflow should record accelerator type and count, wall-clock runtime, average power draw or tracker-based estimates, geographic grid-carbon intensity, and the resulting kgCO₂e for training, hyperparameter search, and inference separately. Software estimators such as CarbonTracker can support runtime accounting, while lifecycle-aware reporting should additionally document embodied-hardware assumptions and data-center context so that operational and embodied emissions are not conflated [217–220]. Journals and funding agencies could further normalize this practice by requiring compute and carbon statements within the methods section or supplementary material.

Several mitigation strategies have been proposed to reduce the carbon footprint of climate AI. Model compression, parameter pruning, quantization, and knowledge distillation enable substantial computational savings while preserving predictive performance [218, 221]. Hardware–model co-optimization further improves energy efficiency by aligning architectural design with computing infrastructure [222]. Carbon-aware scheduling dynamically shifts workloads across regions or time periods with lower carbon intensity [223, 224]. Federated learning architectures additionally reduce centralized computation and data transmission, offering emission efficiency through distributed computation [224]. Structural sustainability measures such as renewable-powered data centers and infrastructure-aware orchestration further reduce lifecycle emissions [217, 220].

Without explicit efficiency constraints, large-scale AI training risks undermining climate mitigation goals through excessive energy demand. Climate AI therefore requires “carbon-aware evaluation,” where performance claims are interpreted relative to computational and environmental cost. This includes normalized reporting of energy usage per experiment, carbon intensity of compute location, and efficiency-to-skill ratios. Embedding sustainability constraints in model selection is essential to ensure that AI-driven decarbonization does not depend on carbon-intensive infrastructure.

Sustainable deployment of climate AI requires that performance gains be evaluated in relation to environmental cost. Accuracy improvements alone do not justify ecological impact if achieved through disproportionate energy expenditure. Transparent reporting of compute budgets, energy consumption, and carbon-equivalent emissions should therefore become a standard component of AI–climate evaluation practices [215, 225]. Computational efficiency must be recognized not merely as a systems optimization metric, but as a scientific and ethical responsibility in climate-aware artificial intelligence research.

6.4 Future directions

6.4.1 Priority 1: *physics-guided and hybrid climate AI*

Future research should intensify the integration of physical structure into learning systems through hybrid AI–physical architectures, neural operators, and differentiable prediction frameworks. Key questions include: (i) How can conservation laws and thermodynamic constraints be enforced dynamically during long-horizon forecasting? (ii) What architectures best balance expressiveness and physical consistency for multi-scale geophysical processes? and (iii) How can physics-informed inductive biases improve robustness under climate non-stationarity and regime transitions [52, 120, 122]?

6.4.2 Actionable research steps (for physics-guided and hybrid models)

To operationalize these priorities, future studies should:

- (i) Design benchmark tasks that explicitly test extrapolation under regime shifts and unprecedented extremes;
- (ii) Compare purely data-driven and physics-guided models under identical inputs and validation protocols;
- (iii) Report failure modes alongside accuracy gains, particularly for out-of-distribution scenarios; and
- (iv) Release reproducible training pipelines to enable controlled comparison across architectures.

For PINNs and neural operators specifically, the central scalability challenges include memory growth at higher spatial resolution, stability over long autoregressive horizons, and the computational cost of enforcing physical constraints during training. Benchmark suites should therefore vary spatial scale, forecast horizon, conservation targets, and compute budget so that the effects of physical constraints on fidelity, long-horizon stability, and interpretability can be assessed jointly rather than through accuracy alone [52, 120, 122, 126].

6.4.3 Priority 2: *federated and privacy-preserving climate intelligence*

Operational climate AI increasingly requires collaboration across institutions, countries, and governance regimes without centralizing data. Federated learning (FL) and privacy-enhancing technologies create opportunities for distributed model training across heterogeneous data silos. Open research challenges include: (i) mitigating performance degradation under non-IID climate data, (ii) ensuring secure aggregation under adversarial conditions, and (iii) achieving interoperability between national weather agencies and research institutions [193, 195, 224].

6.4.4 Priority 3: AI-driven digital twins of the earth system

Digital twins offer a pathway for real-time scenario simulation, policy testing, and adaptive climate-risk management. Future work should focus on: (i) scalable neural surrogates for physical solvers, (ii) continuous assimilation of streaming sensor and satellite data, and (iii) validation infrastructures for synchronizing model states with real-world observations. AI-powered digital twins must further address uncertainty propagation and interpretability to support regulatory and policy use [53, 120, 121].

6.4.5 Priority 4: uncertainty-aware and out-of-distribution (OOD) robust modeling

Forecast reliability under climate extremes depends critically on uncertainty quantification and robustness to distributional shift. Research should prioritize mathematically grounded prediction intervals, calibration under non-stationary conditions, and formal OOD testing. Open questions include: (i) How should ensembles, Bayesian networks, and conformal prediction be benchmarked for climate deployment? and (ii) How can uncertainty be translated into actionable risk metrics for early-warning systems [207, 209, 210]?

6.4.6 Priority 5: ethical governance, data sovereignty, and responsible AI

Future research must extend beyond algorithmic design toward inclusive governance models. Priority areas include: (i) enforcing data sovereignty principles in climate datasets, (ii) mitigating algorithmic bias against underserved regions, and (iii) integrating participatory design into community-scale AI systems. Ethical AI is not auxiliary but fundamental to cloud-based climate intelligence deployed at global scale [119, 193].

6.4.7 Priority 6: Sustainable and carbon-aware model development

Climate AI systems must be held to efficiency standards consistent with their mission. Key directions include energy-normalized benchmarking, carbon-aware workload orchestration, and lifecycle cost modeling of AI pipelines. Future research should formulate compute-efficiency metrics alongside skill scores to prevent environmentally unsustainable algorithm selection [217, 218, 220].

Future advances will hinge on coordinated development across model design, operational integration, and governance frameworks. Bridging physics, data science, and policy is essential for transitioning from experimental models to decision-grade climate AI services. Robust deployment requires open benchmarks, transparent reporting of computational cost, and interdisciplinary collaboration across geoscience, engineering, and public policy.

Decision-ready climate intelligence will further depend on explainable interfaces, uncertainty-aware outputs, and operational validation pipelines. Integrating satellites, in-situ sensors, and Internet-of-Things (IoT) infrastructures into continuously updated digital twins offers a promising direction for real-time risk analytics and long-term resilience planning.

6.5 Future work: quantitative transparency audit

Building on the domain-level compliance estimates reported in Table 14, a natural extension of this review is a fully standardized, paper-by-paper transparency audit. Such an effort would require harmonized operational definitions for code and weight

availability, uncertainty documentation, and energy or CO₂ footprint reporting, as well as consistent annotation across heterogeneous publication formats. We therefore identify this as a priority direction for future work and community meta-research to enable precise, reproducible benchmarking of openness and environmental accountability in climate AI.

7 Conclusion

This review synthesized developments in AI for climate science across five domains: climate modeling, extreme events, renewable energy, emissions monitoring, and cross-domain adaptation. Across tasks such as nowcasting, downscaling, bias correction, and anomaly detection, data-driven and hybrid approaches increasingly complement operational NWP systems. The strongest gains arise when learning systems are constrained by physical structure, linking predictive skill to mechanistic consistency rather than statistical fit alone.

Beyond summarizing applications, this study introduced a review framework that integrates PRISMA screening with a study-quality checklist to shift attention from headline accuracy to evaluation credibility. We show that many reported gains rely on weak validation designs, limited treatment of out-of-distribution behavior, and incomplete uncertainty reporting. Openness remains uneven: code, trained weights, and energy/CO₂ footprints are often unavailable, limiting reproducibility and environmental accountability.

The path toward decision-grade climate AI requires methodological consolidation. Priorities include physics-guided architectures (e.g., PINNs and neural operators), uncertainty-aware forecasting, regime-aware validation, and standardized reporting of baselines, calibration, and compute footprint. Equally critical are open benchmarks, shared model repositories, and carbon-aware evaluation practices to align innovation with sustainability goals.

Ultimately, climate AI will achieve operational impact only through integration with physical understanding, transparent evaluation, and responsible governance. Progress depends not merely on larger models but on reproducible science, energy-aware computation, and inclusive deployment that serves climate-vulnerable communities.

Author contributions

Essam H. Houssein: supervising, methodology, formal analysis, writing—review and editing. Mahmoud Dirar: Methodology, conceptualization, visualization, data curation, resources, writing—review and editing. Abdelmgeid A. Ali: Supervision. Waleed M. Mohamed: Methodology, formal analysis, investigation, writing—review and editing. Kadry Hamed: Methodology, validation, writing—review and editing. All authors read and approved the final paper.

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